CN Practical 8: Create a Network Based on IPv6 Address Using a Simulator

Aim: To create and configure a network using IPv6 addresses in a network simulator.

Theory:

Internet Protocol Version 6 (IPv6) is the latest version of the Internet Protocol developed to overcome the limitations of IPv4 address availability. IPv6 uses a 128-bit address, represented by eight groups of hexadecimal digits separated by colons. For example, 2001:DB8:1::1 is a valid IPv6 address, where consecutive zero groups have been compressed using double-colon notation.

IPv6 supports a very large address space, automatic address configuration and improved network addressing capabilities. In a network simulator such as Cisco Packet Tracer, IPv6 addresses can be assigned to computers and routers. Devices configured with compatible addresses and prefixes can communicate with each other.

Example IPv6 configuration:

Device	IPv6 address	Prefix length
PC1	2001:DB8:1::1	64
PC2	2001:DB8:1::2	64

Network diagram:

```
      PC1 ---------------- PC2
2001:DB8:1::1        2001:DB8:1::2
            Ethernet
```

Procedure:

1. Open Cisco Packet Tracer.
2. Place two PCs in the workspace.
3. Connect both PCs using a suitable Ethernet connection.
4. Open PC1 and select Desktop, followed by IP Configuration.
5. Select IPv6 configuration.
6. Enter 2001:DB8:1::1 as the IPv6 address and set the prefix length to 64.
7. Open PC2's IPv6 configuration.
8. Enter 2001:DB8:1::2 and set the prefix length to 64.
9. Ensure both devices use the same IPv6 prefix.
10. Open the command prompt on PC1.
11. Execute ping 2001:DB8:1::2.
12. Observe the response and verify the connectivity.

Observation:

Both computers were configured with IPv6 addresses belonging to the same /64 subnet. The ping command was used to test communication between the devices. Successful replies indicated that the IPv6 configuration and network connection were functioning correctly.

Conclusion:

An IPv6 network was created and configured using Cisco Packet Tracer. The IPv6 addressing format, prefix length and connectivity testing procedure were studied. Communication between the configured devices was verified using the ping command.